ASSIGNMENT No. 05

One-Dimensional Arrays: *Write programs to perform array operations including insertion, deletion, searching, finding largest/smallest element, frequency count, duplicate removal, and array rotation.*

1. To reverse an array of int/float.
2. To insert an element at a given location of an array.
3. Delete a given element in an array.
4. Linear search for elements.
5. Binary search for an element (which one among linear or binary search is better and why).
6. Sum and average of array elements.
7. Merge two arrays.
8. Find largest and smallest in array.
9. Union and intersection of elements of two arrays.
10. Remove duplicates in an array.
11. To add, multiply and transpose of a matrix.
12. Sort the array elements in ascending order.
13. Rotate array elements (123,231,312) or (abcd, bcda, cdab, dabc).
14. Find mean, median, mode of an array.
15. Use functions, loops and arrays to convert a decimal number into (a) binary (b) octal (c) hexadecimal.

16. Write a program that takes an array of integers and a number n, then shifts all elements of the array to the left by n positions. The elements shifted out from the beginning are discarded, and the vacant positions at the end are filled with -1.
    Example:
    Input:
    Size of array = 5
    Elements = 10 20 30 40 50
    Number of positions to shift = 2
    Output:
    Array after left shifting by 2 positions:
    30 40 50 -1 -1

17. Write a program that takes a one-dimensional array of integers as input and calculates both the mean (average) and the variance of the elements. Display both results rounded to two decimal places.
    Hint: To calculate variance:
    1. Compute the mean of the elements.
    2. For each element, find the squared difference from the mean.
    3. Find the average of these squared differences.

18. Write a program to solve the “Two Sum” problem: find two numbers in an array whose sum equals a given target value.